

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 642 Deep Reinforcement Learning — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Sutton & Barto — Reinforcement Learning: An Introduction, 2nd edition (free PDF)	The primary source, free from the authors. Chapters 2, 5–7, 9–13 are Modules 02–06; chapters 3–4 were CSCE 625 Module 10, which is why the two courses share one book.
OpenAI Spinning Up in Deep RL (free)	The best free practical introduction — clean reference implementations of the Module 05–07 algorithms, with the derivations and an unusually honest discussion of what is hard.
Berkeley CS285 — Deep Reinforcement Learning (free lectures and homework)	The graduate course this one follows most closely. Full video lectures and assignments; Modules 05–09 track it.
Henderson et al. — Deep Reinforcement Learning That Matters (free)	Module 12's source. The seed-variance measurements that should change how you read every other paper in the field.
Agarwal et al. — Deep RL at the Edge of the Statistical Precipice (free)	How to report reinforcement learning results properly — Module 12's recommended practice, with the statistical argument and free tooling.
Gymnasium, Stable-Baselines3, and CleanRL (free)	The working toolkit. CleanRL's single-file implementations are the ones to read — every algorithm in one file, with the implementation details that papers omit.

Module-by-module map

Module 01 — From Planning to Learning

Resource	What it covers for this module
Sutton & Barto — Reinforcement Learning, chapter 1 (free PDF)	The field's own framing , and chapters 3–4 are the CSCE 625 Module 10 material this course builds on.
OpenAI Spinning Up — Introduction and Key Concepts (free)	The clearest statement of §1's framing available , with the taxonomy of §3's table.
Berkeley CS285 — lecture 1 and the course overview (free)	The graduate framing , including an honest treatment of §2's sample-efficiency problem.
Tsividis et al. — Human Learning in Atari (free)	The human comparison of §2, measured — and the analysis of which priors humans bring.

Module 02 — Exploration

Resource	What it covers for this module
Sutton & Barto — chapter 2 (free PDF)	Bandits, §1's strategies, and the regret framing , with the experiments worth reproducing.
Lattimore & Szepesvári — Bandit Algorithms (free PDF)	Free in full, and the reference for §1 and §2 — the UCB and Thompson sampling analyses done properly.
Osband et al. — Deep Exploration via Bootstrapped DQN (free)	§3's corridor argument and the per-episode randomisation answer , from the paper that framed deep exploration clearly.
Burda et al. — Exploration by Random Network Distillation (free)	§4's fix for the noisy-TV problem , with the diagnosis that motivates it.

Module 03 — Learning Values from Experience

Resource	What it covers for this module
Sutton & Barto — chapters 5–7 and 12 (free PDF)	Monte Carlo, temporal difference, n-step, and traces — the reference for this whole module. The cliff-walking example of §2 is figure 6.5.
Watkins & Dayan — Q-learning (free)	The original convergence result — short, and worth reading for what the proof requires.
Hasselt — Double Q-learning (free)	§4's analysis and fix , with the bias demonstrated on a small example.
David Silver — Reinforcement Learning lectures 4–5 (free video)	Model-free prediction and control , and the clearest spoken explanation of §2's distinction.

Module 04 — Deep Q-Networks

Resource	What it covers for this module
Mnih et al. — Human-level control through deep reinforcement learning (free)	The DQN paper. The replay and target-network motivation of §2 is stated here, and the engineering details matter.
Sutton & Barto — chapter 11 (free PDF)	The deadly triad and Baird's counterexample of §1 , developed carefully. This chapter is the one to read slowly.
Hessel et al. — Rainbow: Combining Improvements in Deep Reinforcement Learning (free)	§3's list with the ablation study that says which components actually contribute.
CleanRL — dqn.py and dqn_atari.py (free)	A readable single-file DQN with every implementation detail visible — and the details are where the performance is.

Module 05 — Policy Gradients

Resource	What it covers for this module
Sutton & Barto — chapter 13 (free PDF)	The policy gradient theorem and REINFORCE , with the baseline derivation of §2.
OpenAI Spinning Up — Intro to Policy Optimization (free)	The clearest available derivation of §1 , including the reward-to-go and baseline arguments with the algebra shown.
Williams — Simple statistical gradient-following algorithms (free)	The original REINFORCE paper , and it is clearer than its reputation.
Schulman et al. — High-Dimensional Continuous Control Using Generalized Advantage Estimation (free)	§2's variance reduction taken properly — GAE interpolates the bias-variance trade in the advantage, and it is what every modern implementation uses.

Module 06 — Actor-Critic and Trust Regions

Resource	What it covers for this module
Schulman et al. — Proximal Policy Optimization Algorithms (free)	PPO. Short, and read it alongside §3's detail list rather than alone.
Schulman et al. — Trust Region Policy Optimization (free)	§2's principled version , with the monotone improvement argument that PPO approximates.
Engstrom et al. — Implementation Matters in Deep Policy Gradients (free)	§3's finding, measured — and the paper that should change how you read comparisons in this field.
CleanRL — ppo.py and the PPO implementation-details blog (free)	Every detail from §3, enumerated and benchmarked. The single most useful page in this course.

Module 07 — Continuous Control

Resource	What it covers for this module
Haarnoja et al. — Soft Actor-Critic (free)	§3's method , with the maximum-entropy framing derived and the auto-tuned temperature in the follow-up paper.
Fujimoto et al. — Addressing Function Approximation Error in Actor-Critic Methods (TD3) (free)	§2's three fixes , with the overestimation analysis that motivates all three.
Lillicrap et al. — Continuous control with deep reinforcement learning (DDPG) (free)	§1's deterministic policy gradient — read for the idea, not as a recommendation.
Tobin et al. — Domain Randomization (free)	§4's transfer method , and a clear statement of the reality-gap problem it addresses.

Module 08 — Model-Based Methods

Resource	What it covers for this module
Sutton & Barto — chapter 8 (free PDF)	Dyna and the planning-learning relationship , in the tabular setting where it is clearest.
Janner et al. — When to Trust Your Model: Model-Based Policy Optimization (free)	§2's short-rollout defence , with the analysis that says how short.
Hafner et al. — Dream to Control, and DreamerV3 (free)	§3's latent world model , and DreamerV3 is notable for working across many domains with fixed hyperparameters — which is rare in this field.
Schrittwieser et al. — Mastering Atari, Go, Chess and Shogi by Planning with a Learned Model (MuZero) (free)	§4's method . Read it for the observation that the model need not model the world.

Module 09 — Learning Without Interaction

Resource	What it covers for this module
Levine et al. — Offline Reinforcement Learning: Tutorial, Review, and Perspectives (free)	The reference for §2 and §4 . The extrapolation analysis is the clearest statement of the core failure.
Ross, Gordon & Bagnell — A Reduction of Imitation Learning and Structured Prediction (DAgger) (free)	§1's method , with the quadratic-to-linear bound proved.
Kostrikov, Nair & Levine — Offline RL with Implicit Q-Learning (free)	§2's recommended method , and notably simple for what it achieves.
Fu et al. — D4RL: Datasets for Deep Data-Driven Reinforcement Learning (free)	Standard offline datasets , including the mixed-quality ones that make §3's stitching question concrete.

Module 10 — Reward Design

Resource	What it covers for this module
Krakovna et al. — Specification gaming examples in AI (free, maintained list)	§1's examples and dozens more , collected and sourced. Browse it before designing any reward.
Ng, Harada & Russell — Policy Invariance Under Reward Transformations (free)	§2's theorem , with the telescoping argument. Short and worth reading in full.
Andrychowicz et al. — Hindsight Experience Replay (free)	§3's relabelling trick , which is simple and unreasonably effective on sparse goal-reaching tasks.
Christiano et al. — Deep RL from Human Preferences (free)	§4's method , including the reward-hacking observation and the continued-collection response.

Module 11 — Multiple Agents and Self-Play

Resource	What it covers for this module
Silver et al. — A general reinforcement learning algorithm that masters chess, shogi and Go (AlphaZero) (free)	§2's self-play result , with the curriculum property doing the work.
Vinyals et al. — Grandmaster level in StarCraft II with multi-agent reinforcement learning (free)	§2's league training , and the clearest demonstration of why opponent populations are necessary — and of the compute cost.
Rashid et al. — QMIX (free)	§3's monotone value decomposition , with the consistency argument.
Lanctot et al. — A Unified Game-Theoretic Approach to Multiagent RL (PSRO) (free)	§2's population method , and a good treatment of why non-transitivity demands it.

Module 12 — Reproducibility

Resource	What it covers for this module
Henderson et al. — Deep Reinforcement Learning That Matters (free)	§1's measurements , including the same-algorithm split-group demonstration. Read this before trusting any comparison.
Agarwal et al. — Deep RL at the Edge of the Statistical Precipice (free)	§2's recommended statistics , with the free Rliable library that computes interquartile means and performance profiles.
Engstrom et al. — Implementation Matters in Deep Policy Gradients (free)	§3's first row, measured — the implementation confound isolated from the algorithmic one.
Jordan et al. — Evaluating the Performance of Reinforcement Learning Algorithms (free)	A complete evaluation protocol , including the hyperparameter-effort confound of §3.

Module 13 — Where It Works

Resource	What it covers for this module
Irpan — Deep Reinforcement Learning Doesn't Work Yet (free)	The honest assessment that this module's §2 follows , by a practitioner. Dated in its examples and not in its reasoning.
Dulac-Arnold et al. — Challenges of Real-World Reinforcement Learning (free)	§2's failure conditions, enumerated systematically , with the gap between benchmark and deployment settings made explicit.
Mirhoseini et al. — A graph placement methodology for fast chip design (free)	§1's chip placement row — and worth reading alongside its published critiques, as an exercise in Module 12 §4's questions.
Sutton & Barto — chapters 16–17 (free PDF)	Case studies and frontiers , and a reasonable place to choose what to study next.